Attachment 1
Contract No. 271717-23
Document ID: SOW- 21233
Revision ID: 1
Effective Date: 08/07/2024

Statement of Work

MACHINE LEARNING BASED CORRELATION THE MECHANICAL PROPERTIES OF SUB-SIZED SPECIMENS AND STANDARD-SIZED SPECIMENS

Idaho National Laboratory

The INL is a U.S. Department of Energy National Laboratory operated by Battelle Energy Alliance.

REVISION LOG

Rev.	Date	Affected Pages	Revision Description
0	11/01/2023	All	New Document
1	08/07/2024	Page 6	Update the new deliveries for the new FYs

CONTENTS

1. INTRODUCTION 4

1.1 Background 4

1.2 Purpose / Objectives 4

1.3 Anticipated Benefits 5

2. APPLICABLE CODES AND REFERENCES 5

3. SCOPE 5

3.1 Work to be Performed 5

3.2 Work Excluded 6

3.3 Requirements 6

3.4 Place of Performance 6

3.5 Interfaces 6

3.6 Miscellaneous 6

4. DELIVERABLES 6

5. SCHEDULE AND MILESTONES 6

6. COMPLETION CRITERIA AND FINAL ACCEPTANCE 6

7. APPENDICES 7

8. ATTACHMENTS 7

1. INTRODUCTION

1.1 Background

Battelle Energy Alliance (BEA) is the current management and operating contractor for the Department of Energy (DOE) located at Idaho National Laboratory (INL). Small-scale mechanical testing using sub-sized specimens is critical in the nuclear industry to accelerate irradiation tests, to gain knowledge of new materials during/after irradiation, and to ultimately accelerate materials qualification and deployment. However, the small size of a specimen causes the so-called "size effect", which leads to different material behaviors in the microscale, mesoscale, and macroscale. The goal of this project is to correlate the mechanical properties of sub-sized specimens and those of standard-sized specimens using machine learning (ML). The objectives of this project are to: (1) build a database of materials properties, e.g., Charpy impact toughness, tensile, and creep properties, including detailed pedigree information for the selected materials. (2) identify and evaluate existing methods for properties prediction and develop ML methodologies that can be used for more accurate properties prediction under different test factors; (3) develop a correlation of sub-sized and standard-sized specimens under different conditions; (4) investigate the importance of different test factors to the corresponding mechanical properties, and eventually guide test matrix design and accelerate new material development.

The machine intelligence and data analytics lab at University of Idaho can enhance the applications of artificial intelligence (AI) and machine learning (ML) techniques to nuclear material data and work with BEA for the LDRD project. The subcontractor from the lab has demonstrated and applied the physic ML models to structure material creep fatigue property study. Expert consulting services in the areas of data analysis and machine learning modeling, especially working on the limited unbalanced public data, are required to pursue the outcomes of the project. The results can be used to correlate the mechanical properties between sub-sized and standard-sized specimens based on ML models to accelerate materials qualification and deployment, which is extremely important for the development and qualification of structural materials for Generation IV nuclear energy systems where the long-term high temperature property is required, and accurate empirical extrapolation methods are needed.

1.2 Purpose / Objectives

The purpose of this statement of work (SOW) is to facilitate collaboration between University of Idaho and BEA to provide good technical quality products for correlating the sub-size and standard-size effects with the corresponding structural material's mechanical properties. The subcontractor is deemed qualified as an expert in the area of AI/ML to provide advice and assistance to prepare the products for the project. Furthermore, this project will facilitate

workforce development of graduate and postdoctoral researchers from University of Idaho and/or BEA to complete their studies.

The objective of this subcontract is to involve ML experts with nuclear domain knowledge to work together with BEA to achieve the project goals.

1.3 Anticipated Benefits

This statement of work (SOW) aims to further enhance the collaborations between University of Idaho and BEA through the mentorship of graduate and/or postdoctoral researchers, further fostering workforce development through potential candidates for full-time staff researcher positions at INL. Furthermore, interactions with University of Idaho enhances technical capabilities of BEA to rapidly assess nuclear structural materials for applications in advanced nuclear reactors, particularly in relation to size effects on structure material. The subcontractor will provide great technical support to project BEA by identifying and evaluating advanced computational tools and techniques for nuclear structural material design and testing.

2. APPLICABLE CODES AND REFERENCES

N/A

3. SCOPE

3.1 Work to be Performed

Subcontractor shall provide assistance to BEA for the following tasks and other related studies as directed by technical point of contact (TPOC):

- Building databases for the studied material and mechanical properties, which will feed into the ML models.
- Reviewing current statistic and AI/ML models for the corresponding property study
- Designing new models to identify the key irradiation factors impacting the corresponding property and predict the property performance given new irradiation conditions.
- Providing user friendly interface or manual documents for the BEA to use the models
- Collaboration with paper publication.
- The newly developed code and/or software is publicly accessible through GitHub or other websites and the ownership belongs to BEA and University of Idaho.

3.2 Work Excluded

N/A

3.3 Requirements

The assigned tasks must be completed as scheduled and to the satisfaction of the BEA Technical Point of Contact (TPOC).

3.4 Place of Performance

University of Idaho and Idaho National Laboratory, if needed.

3.5 Interfaces

3.6 Miscellaneous

N/A

4. DELIVERABLES

The Subcontractor shall provide the products in the form of journal papers, reports, white papers, and guidance documents, as well as recommendations, technical judgements, and insights for the tasks in this project.

The delivery of the subcontract shall be a dataset of tensile, Charpy impact, and creep property of candidate materials in required format and one journal draft with title "Correlation of tensile properties between sub-sized and standard-sized samples via machine learning based model for nuclear materials."

5. SCHEDULE AND MILESTONES

The literature work shall start at the effective date.

The dataset of tensile property shall be established.

The draft of the journal paper shall be sent to the BEA TPOC.

6. COMPLETION CRITERIA AND FINAL ACCEPTANCE

Assigned tasks must be completed to the satisfaction of the BEA TPOC. Receiving of deliverables and approval from the BEA TPOC is required for completion and final acceptance. Acceptance shall not be unreasonably withheld and shall be determined based on the requirements stated in this Contract Release (including all applicable documents).

7. APPENDICES

N/A

8. ATTACHMENTS

N/A